

PENETRATION TESTING REPORT

FOOTPRINTING, RECONNAISSANCE & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY & ETHICAL HACKING | NETWORKWALKS

Field	Details
Pentester Name	Ishanya Jha
Program/Batch	B083-Networkwalks
Program	Cybersecurity & Ethical Hacking Internship
Organization	Networkwalks
Report	W2-PM-FINAL
Week	Week 2
Date	19 September 2026
Modules Completed	W2-PM1, W2-PM2, W2-PM3, W2-PM4, W2-PM5
Client/Target	Networkwalks + Own Local LAN
Permission	Yes, within assigned/authorized scope
Phase 1	Reconnaissance & Footprinting: Completed
Phase 2	Scanning & Network Discovery: Completed

1. LIABILITY DISCLAIMER

All activities documented in this report were performed as part of the assigned educational cybersecurity internship exercises and within the applicable authorized scope.

The activities were conducted for educational and security-research purposes. No exploitation, unauthorized access, privilege escalation, denial-of-service activity, brute-force activity, or intentional modification, deletion, or destruction of data was performed.

Reconnaissance activities involving Networkwalks were limited to the assigned scope. Zenmap network discovery was performed on the tester's own local LAN.

Security testing must only be performed against systems and networks that are owned by the tester or where appropriate authorization has been established.

2. INTRODUCTION

This report documents the completion of Week 2 of the Networkwalks Cybersecurity & Ethical Hacking Internship.

The practical work covered:

- Phase 1: Reconnaissance & Footprinting
- Phase 2: Scanning & Network Discovery

The five completed project modules were:

- W2-PM1: Footprinting using multiple Kali Linux tools
- W2-PM2: Footprinting & Reconnaissance using GHDB
- W2-PM3: Footprinting using Maltego
- W2-PM4: Footprinting & Reconnaissance using theHarvester
- W2-PM5: Network Scanning using Zenmap

The exercises demonstrated how publicly available information can be collected during reconnaissance and how authorized network scanning can identify active devices within a local network.

Evidence has been placed directly under the relevant task throughout this report.

3. SCOPE AND AUTHORIZATION

3.1 Networkwalks Reconnaissance

The assigned Networkwalks domain used for the reconnaissance exercises was: networkwalks.com

The following activities were performed:

- WHOIS
- WhatWeb

- NSLookup
- cURL
- Wafw00f
- DNSRecon
- GHDB reconnaissance
- Maltego
- theHarvester

These activities were conducted within the assigned educational scope.

3.2 Local Network Scanning

The Zenmap exercise was performed against the tester's own local LAN.

The scope included:

- Local IP identification
- Subnet identification
- Live-host discovery
- IP address identification
- MAC address identification
- Network topology generation

No external network was intentionally scanned as part of the Zenmap practical.

4. TOOLS USED

Tool	Purpose
Kali Linux	Reconnaissance environment
WHOIS	Domain registration information
WhatWeb	Web technology fingerprinting
NSLookup	DNS resolution
cURL	HTTP response inspection
Wafw00f	WAF identification
DNSRecon	DNS enumeration
GHDB	Search-engine reconnaissance
Maltego	Relationship and domain reconnaissance
theHarvester	Public-source information gathering
Zenmap	Local network discovery
Windows CMD	Local network configuration

5. ACTIVITIES PERFORMED

5.1 W2-PM1: FOOTPRINTING & RECONNAISSANCE

W2-PM1 consisted of six reconnaissance tasks performed against the assigned Networkwalks domain.

5.1.1 Task 1: WHOIS

WHOIS was used to obtain publicly available domain-registration information for networkwalks.com.

Observed Results

Domain: NETWORKWALKS.COM

Registrar: GoDaddy.com, LLC

Creation Date: 6 November 2019

Updated Date: 12 November 2025

Expiry Date: 6 November 2027

Name Server: NS6135.HOSTGATOR.COM

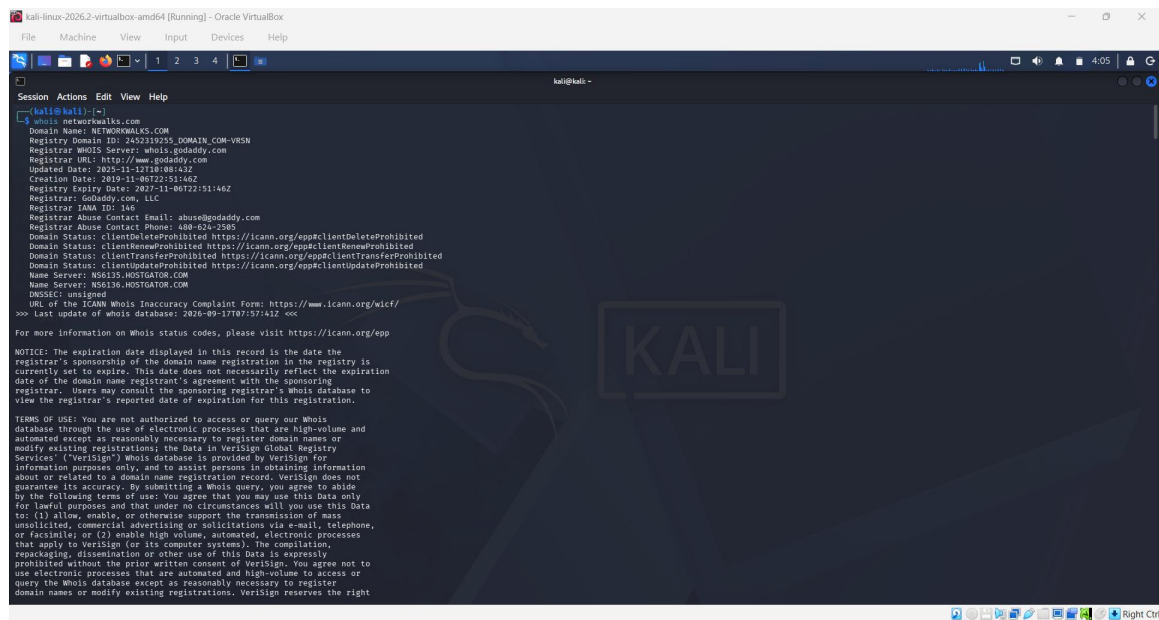
Name Server: NS6136.HOSTGATOR.COM

DNSSEC: Unsigned

The collected output also contained several client protection status codes.

No registrant identity was returned in the collected result.

Evidence



```
kali@kali:~$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452239253_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T18:08:41Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2025-11-12T17:17:12Z

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the data in VeriSign Global Registry
Services' (VeriSign) Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this data only
for lawful purposes and that under no circumstances will you use this data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
```

Figure 1. WHOIS command output
Supporting output: whois_networkwalks.txt

Security Relevance

WHOIS information can help establish an initial profile of a domain's registration and DNS infrastructure.

5.1.2 Task 2: WhatWeb

WhatWeb was used to fingerprint publicly identifiable web technologies associated with the Networkwalks website.

Observed Results

The collected output identified:

- Apache
- WordPress
- WordPress Download Manager
- Bootstrap
- jQuery
- Google Tag Manager
- HTML5
- Open Graph
- Website IP information
- Page title: Networkwalks Academy

The website returned an HTTP redirect to HTTPS and an HTTPS 200 response.

Evidence

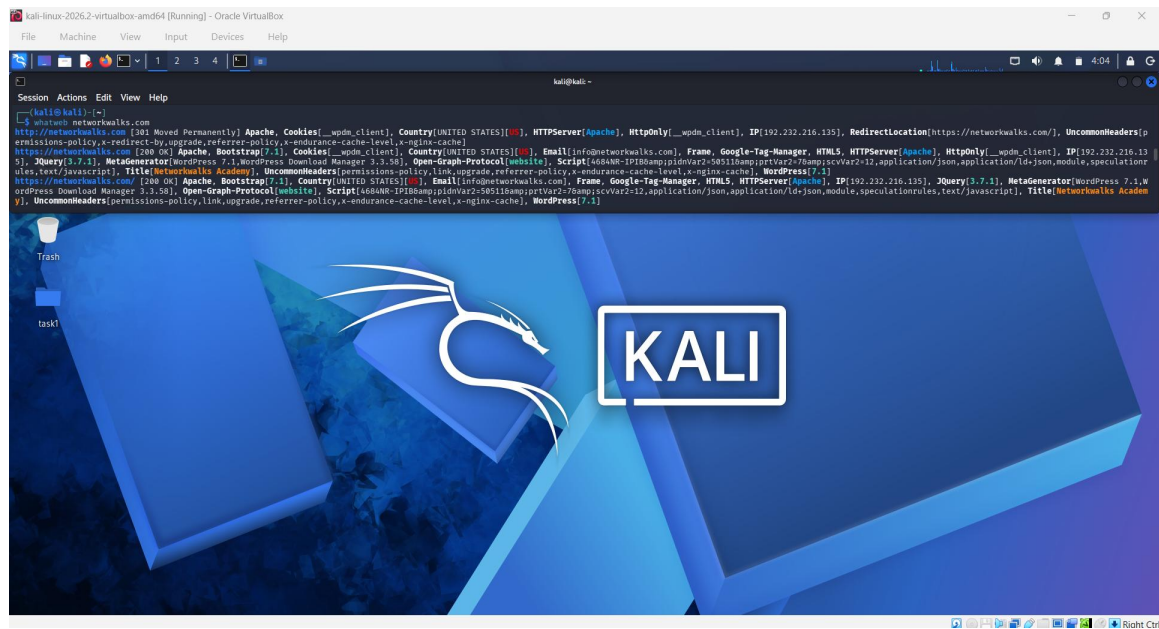


Figure 2. WhatWeb technology fingerprinting
Supporting output: whatweb_networkwalks.txt

Security Relevance

Technology fingerprinting can assist defenders in understanding what software and frameworks are publicly identifiable.

A detected technology or version does not by itself establish a vulnerability.

5.1.3 Task 3: NSLookup

NSLookup was used to resolve the Networkwalks domain.

Observed Results

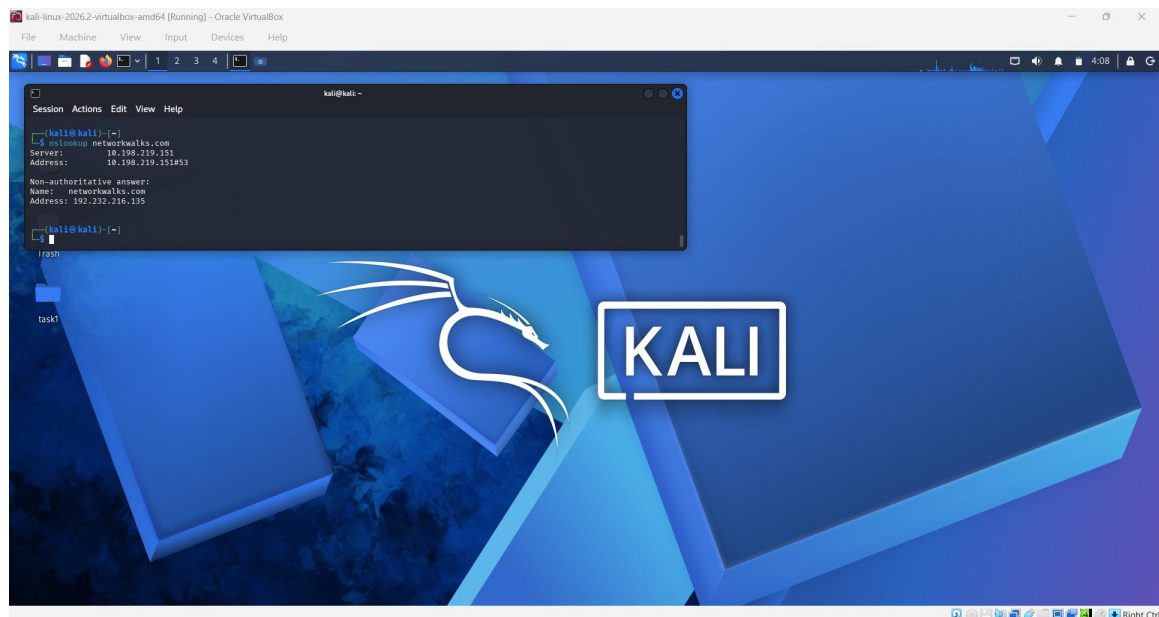
DNS Server: 10.198.219.151

Port: 53

Response: Non-authoritative

Resolved IP: 192.232.216.135

Evidence

A screenshot of a Kali Linux virtual machine window. The terminal window is open, showing the output of the 'nslookup networkwalks.com' command. The output indicates a non-authoritative answer from a DNS server at 10.198.219.151, resolving the domain to the IP address 192.232.216.135. The terminal window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The background of the VM window shows the Kali Linux desktop with a blue and purple geometric pattern and the 'KALI' logo.

```
kali@kali:~$ nslookup networkwalks.com
Server: 10.198.219.151
Address: 10.198.219.151#53

Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135

kali@kali:~$
```

Figure 3. NSLookup domain-resolution output

Supporting output: nslookup_networkwalks.txt

Security Relevance

DNS resolution provides information about the network location of a public service and contributes to infrastructure profiling.

5.1.4 Task 4: cURL

cURL was used to inspect HTTP response headers from the target website.

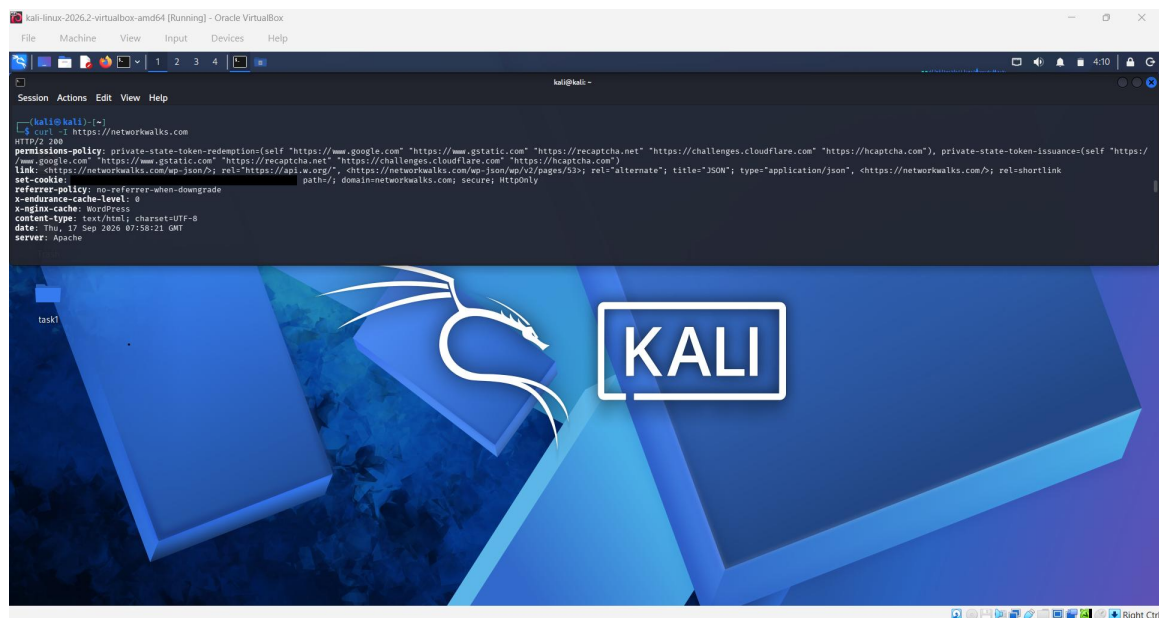
Observed Results

The collected output included:

- HTTP/2 200 response
- Apache server information
- HTML content type
- Permissions-Policy
- WordPress REST API references
- Referrer-Policy
- Cache-related information
- Cookie configuration

Sensitive cookie values were not reproduced in this report.

Evidence



```
kali@kali:~$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/?rel="https://api.w.org">; <https://networkwalks.com/wp-json/wp/v2/pages/33>; rel="alternate"; title="350M"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie:
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Thu, 17 Sep 2026 07:58:21 GMT
server: Apache
```

Figure 4. cURL HTTP response headers
Supporting output: curl_networkwalks.txt

Security Relevance

HTTP response headers can reveal information about an application's technology and configuration. Organizations should review exposed headers and remove unnecessary technical information where appropriate.

5.1.5 Task 5: Wafw00f

Wafw00f was used to identify the Web Application Firewall protecting the website.

Observed Result

Detected WAF: ModSecurity (SpiderLabs)
The scan completed using two requests.

Evidence

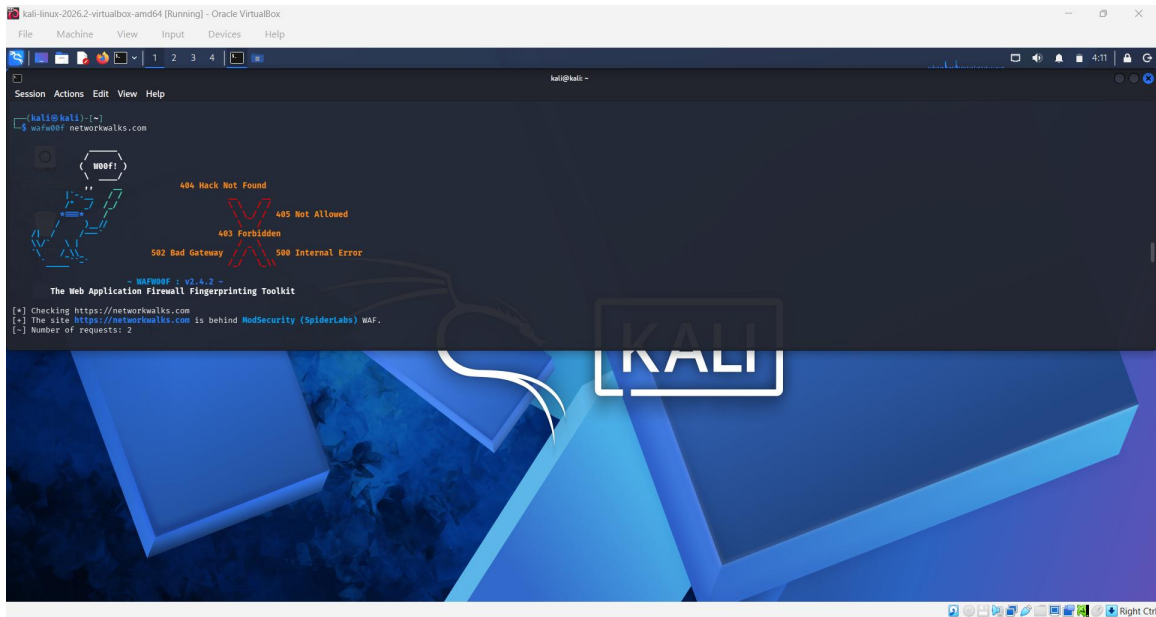


Figure 5. Wafw00f WAF detection
Supporting output: wafw00f_networkwalks.txt

Security Relevance

WAF identification reveals part of an application's defensive architecture.
The identification of a WAF does not indicate that the WAF is ineffective or vulnerable.

5.1.6 Task 6: DNSRecon

DNSRecon was used to enumerate publicly available DNS information.

Observed Results

The collected results included:

- SOA record
- NS records
- MX record
- A record
- TXT records
- SPF record
- SRV records

- DNS software information
- DNSSEC query information

Evidence

```

kali@kali:~$ dnsrecon -u networkwalks.com
2026-09-17T03:58:48.584531-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-17T03:58:48.586558-0400 INFO std: Performing General Enumeration against: networkwalks.com
2026-09-17T03:58:50.414258-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-09-17T03:58:50.604548-0400 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-09-17T03:58:51.281499-0400 INFO NS ns6135.hostgator.com 192.232.216.135
2026-09-17T03:58:51.980772-0400 INFO Bind Version for 192.232.216.131 "9.10.23-RM"
2026-09-17T03:58:51.988722-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-09-17T03:58:54.612808-0400 INFO Bind Version for 50.87.144.87 "9.10.23-RM"
2026-09-17T03:58:54.715449-0400 INFO MX mail.networkwalks.com 192.232.216.135
2026-09-17T03:58:54.765572-0400 INFO A networkwalks.com 192.232.216.135
2026-09-17T03:58:56.766608-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmHmW3XemlZDmV4q75X-1j-mjFeg-UyVI
2026-09-17T03:58:56.766628-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com -all
2026-09-17T03:58:59.377631-0400 INFO Enumerating SRV Records
2026-09-17T03:59:04.817937-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443
2026-09-17T03:59:04.818120-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443
2026-09-17T03:59:04.818372-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443
2026-09-17T03:59:04.818208-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443
2026-09-17T03:59:04.818226-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443
2026-09-17T03:59:04.818249-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443
2026-09-17T03:59:04.818274-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 443
2026-09-17T03:59:04.819294-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443
2026-09-17T03:59:04.819448-0400 INFO SRV 8 Records Found
2026-09-17T03:59:04.819751-0400 INFO Completed enumeration for domain: networkwalks.com

```

Figure 6. DNSRecon enumeration output
Supporting output: dnsrecon_networkwalks.txt

Security Relevance

DNS information can provide a broader picture of an organization's infrastructure. Periodic DNS review can help ensure that unnecessary records and services are not exposed.

5.2 W2-PM2: FOOTPRINTING & RECONNAISSANCE WITH GHDB

Date Added	Dork	Category	Author
2023-11-07	intitle:"Webcam" inurl:WebCam.htm	Various Online Devices	s Thakur
2023-03-14	Google Dork : intitle:"Toshiba Network Camera"	Various Online Devices	Shilpa Shaji
2023-02-27	intitle:"index of /cam/"	Files Containing Juicy Info	Sanu Jose M
2023-01-31	intitle:"index of /webcam/"	Files Containing Juicy Info	Shuvrosayar Das
2022-07-25	intitle:"Device(IP CAMERA)" "language" -com:net	Pages Containing Login Portals	s Thakur
2022-07-25	intitle:"NoVus IP camera" -com	Various Online Devices	s Thakur
2022-07-20	intitle:"Network Camera" inurl:/main.cgi	Various Online Devices	s Thakur
2022-06-17	inurl:webcam site:skylinewebcams.com inurl:roma	Various Online Devices	Simone Gasparato
2021-11-16	intitle:"Login" intext:"cam"	Pages Containing Login Portals	Yash Singh
2021-11-16	intext:"Real-time IP Camera Monitoring System" intext:"ActiveX Mode (For IE Browser)"	Various Online Devices	Yash Singh
2021-11-12	intitle:"Login" intext:"camera"	Pages Containing Login Portals	Naveen Venugopal
2021-11-09	intitle:"webcamXP" inurl:8080	Various Online Devices	Krishna Agarwal
2021-11-09	intitle:"Index of" "DCIM/camera"	Files Containing Juicy Info	Krishna Agarwal
2021-10-28	intitle:ip camera login page	Pages Containing Login Portals	jawhar milkan

5.2.1 Task 1: Security Camera Search

The task involved using GHDB-style search queries to identify publicly indexed webcam-related resources.

The collected results were documented as part of the practical exercise.

No attempt was made to bypass authentication or access controls.

Live-camera screenshots were not reproduced in this report to avoid unnecessarily capturing or redistributing identifiable individuals or sensitive surveillance content.

Evidence

Tasks

Task1 Find 10x live vulnerable security camera links that are exposed & accessible from Internet. List them in below table along with their relevant dork that you have used.

No.	Link	Relevant Dork	Username /Password (if any)
1	http://122.116.41.8:8080/	intitle:"webcamXP" inurl:8080	---
2	https://www.lmc.edu/webcam.htm	intitle:"Webcam" inurl:WebCam.htm	---
3	https://luwebcam.towson.edu/index.html	intitle:"Webcam" inurl:WebCam.htm	---
4	https://www.nps.gov/subjects/air/webcams.htm?site=dena	intitle:"Webcam" inurl:WebCam.htm	---
5	https://pixelhealbandcamp.com/album/intitle-toshiba-network-camera-user-login	Google Dork : intitle:"Toshiba Network Camera"	Cookies/ login required
6	http://99.114.240.169:8080/	intitle:"Webcam" inurl:WebCam.htm	---
7	https://www.shodan.io/search?query=webcam	intitle:"Webcam" inurl:WebCam.htm	Login required
8	https://www.skylinewebcams.com/webcam/italia/lazio/roma/piazza-di-spagna.html	inurl:webcam site:skylinewebcams.com inurl:roma	----
9	https://www.skylinewebcams.com/webcam/italia/lazio/roma/fontana-di-trevi.html	inurl:webcam site:skylinewebcams.com inurl:roma	---
10	https://www.skylinewebcams.com/en/webcam/italia/lazio/roma/roma-colosseo.html	inurl:webcam site:skylinewebcams.com inurl:roma	---

Figure 7. GHDB reconnaissance result
Supporting output: ghdb_live_cam_result.txt

Security Relevance

Search engines can unintentionally index sensitive resources. Organizations should monitor publicly indexed content and ensure that administrative interfaces and private resources are properly protected.

5.2.2 Task 2: Mathematics PDF Listings

The second task focused on identifying publicly indexed mathematics PDF directories.

The search query used was:

intitle:index.of "parent directory" mathematics pdf

Ten results were documented as part of the practical exercise.

Evidence

Task2 Find 10x listings which contain downloadable mathematics ebooks in PDF format:

No.	Link	Relevant Dork	Username /Password (if any)
1	http://inis.lnr.ru/sl/vol2/Mathematics/Math_Encyclopedia/Pdf/	intitle:index.of "parent directory" mathematics pdf	---
2	http://erewhon.superkuh.com/library/Math/	intitle:index.of "parent directory" mathematics pdf	---
3	https://sajapuriacollege.ac.in/pdf/pdf/MATHEMATICS/	intitle:index.of "parent directory" mathematics pdf	---
4	https://www.unm.edu/~megrad/Math/	intitle:index.of "parent directory" mathematics pdf	---
5	https://www.netlib.org/math/docpdf/	intitle:index.of "parent directory" mathematics pdf	---
6	https://www.wvfa.org/pdf/project/LearningTree/	intitle:index.of "parent directory" mathematics pdf	---
7	https://ochicken.neilibrary/Mathematics/	intitle:index.of "parent directory" mathematics pdf	---
8	https://education.qikonda.org.uk/Maths?SD	intitle:index.of "parent directory" mathematics pdf	---
9	https://www.learn-fo.com/FYUG%20in%20mathematics%20solutions/?SD	intitle:index.of "parent directory" mathematics pdf	---
10	https://math.dartmouth.edu/~carlp/PDF/	intitle:index.of "parent directory" mathematics pdf	---

Figure 8. GHDB mathematics PDF search evidence

Supporting file:

W2-PM2 -Footprinting-&-Reconnaissance-with-GHDB-TABLE.pdf

Security Relevance

This exercise demonstrated how search operators can identify publicly indexed files and directories.

Organizations should periodically review what information and files are discoverable through search engines.

5.3 W2-PM3: FOOTPRINTING WITH MALTEGO

5.3.1 Task 1: Download, Installation and Configuration

Maltego was installed and configured on a Windows computer for the practical exercise. A Domain entity was added to the workspace as required by the module.

Evidence

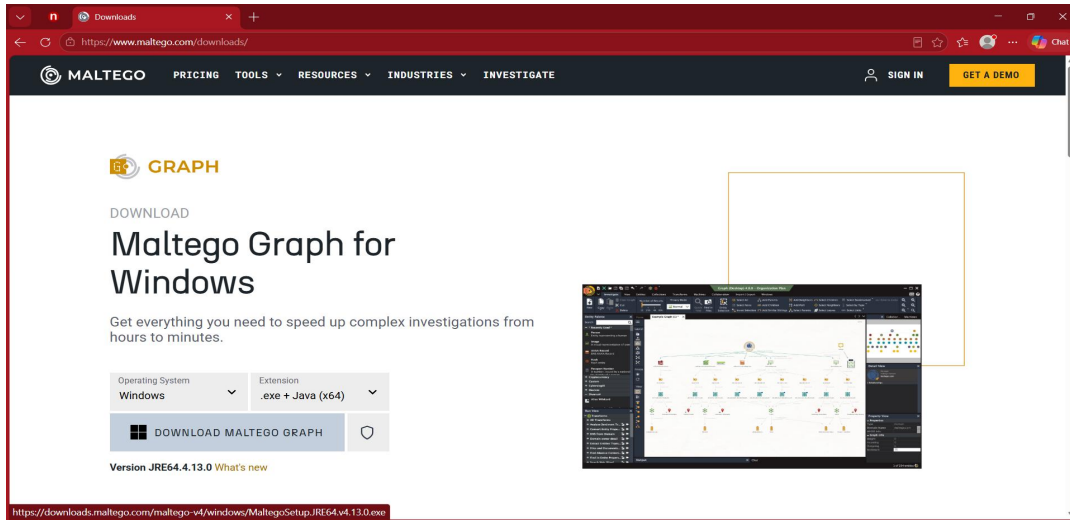


Figure 9. Maltego initial workspace

5.3.2 Task 2: Domain-Based Email Reconnaissance

The authorized target domain used for the exercise was: networkwalks.com .The practical used email-related reconnaissance transforms.

Step 1: Maltego Workspace

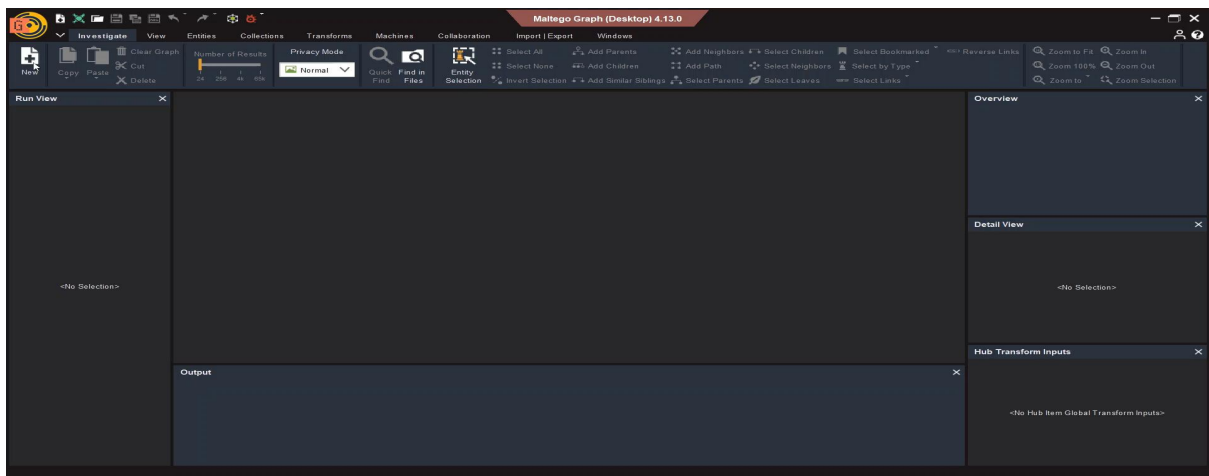


Figure 10. Maltego workspace for Task 2

Step 2: Domain Entity

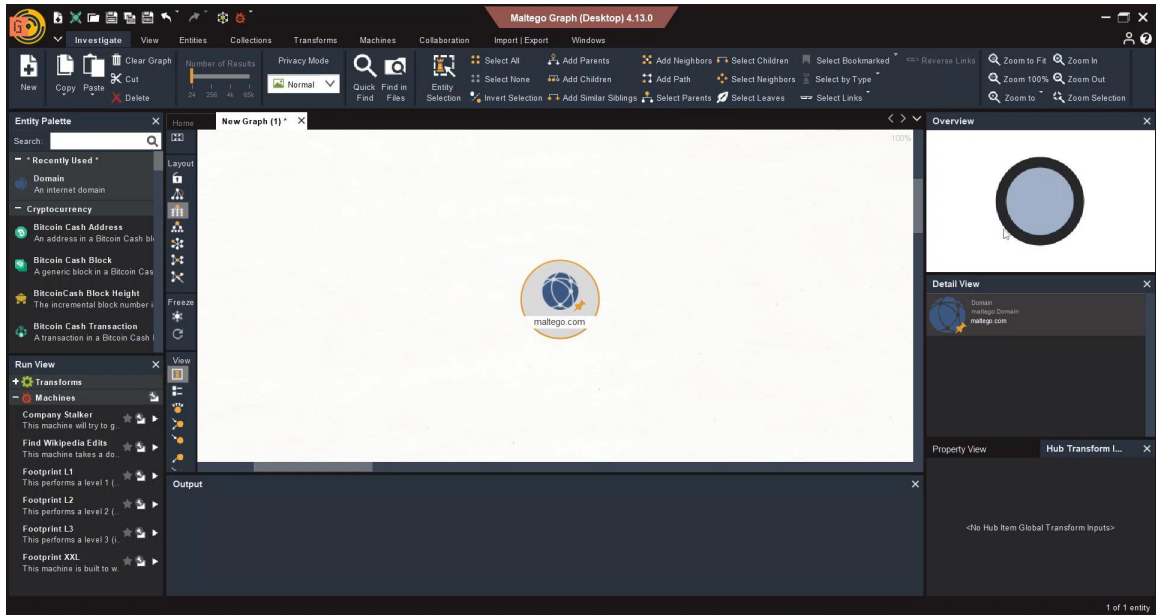


Figure 11. Domain entity configuration

Step 3: Target Configuration

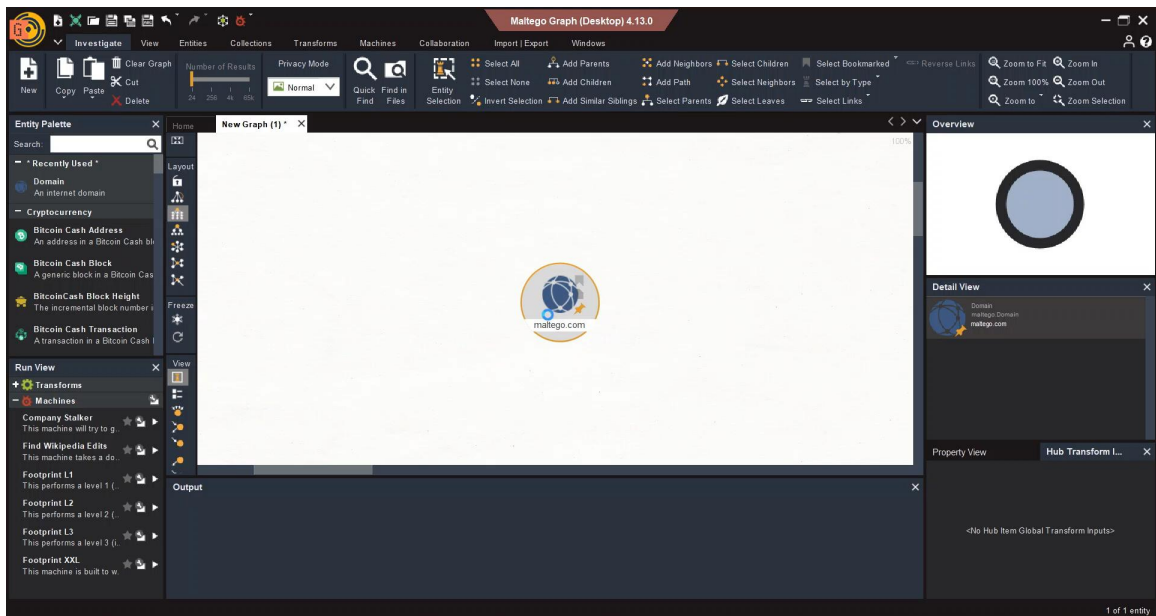


Figure 12. Target domain configuration

Step 4: Email Reconnaissance Transforms

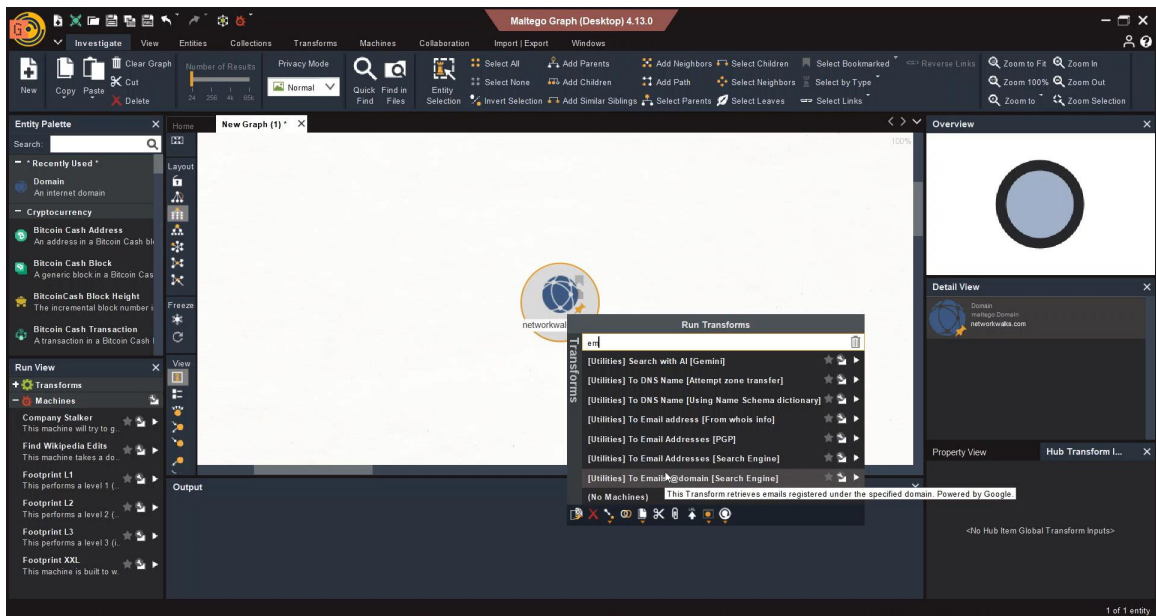


Figure 13. Email-related reconnaissance transforms

Step 5: Output

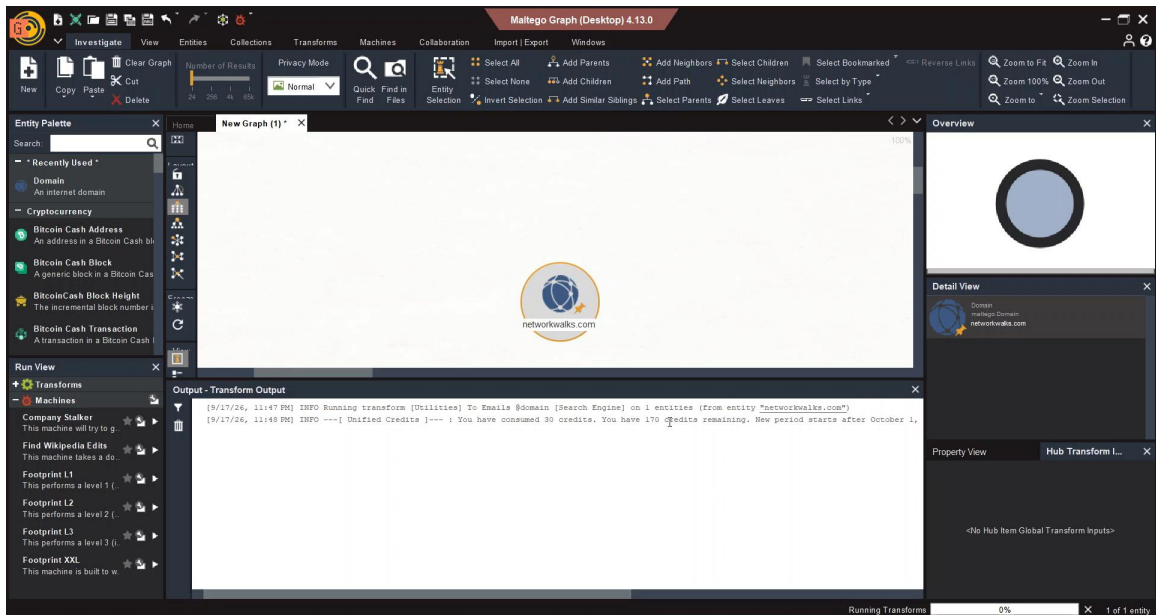


Figure 14. Maltego reconnaissance output
Supporting output:

Task02-output.txtW2-PM3-MALTEGO

[9/17/26, 11:47 PM] INFO Running transform [Utilities] To Email Addresses [Search Engine] on 1 entities (from entity "networkwalks.com")

[9/17/26, 11:48 PM] INFO ---[Unified Credits]--- :You have consumed 30 credits.You have 170 credits remaining.New period starts after October 1, 2026, 12:00:00 AM Z.

Transform: Utilities → To Email Addresses [Search Engine]

Entity: networkwalks.com

Status: Transform initiated successfully.

=====

Transform Evidence

The collected log documented execution of transforms including:

- Utilities → To Email Addresses [Search Engine]
- Utilities → To Email Addresses [PGP]
- Utilities → To Emails @domain [Search Engine]

The transform logs demonstrate that the reconnaissance operations were executed. They should not be interpreted as proof that every transform returned valid email addresses.

Security Relevance

Maltego demonstrates how publicly available information can be collected and correlated to create a broader organizational profile.

5.4 W2-PM4: FOOTPRINTING & RECONNAISSANCE WITH THEHARVESTER

W2-PM4 focused on public-source reconnaissance using theHarvester.

The assigned target domain was:
microsoft.com

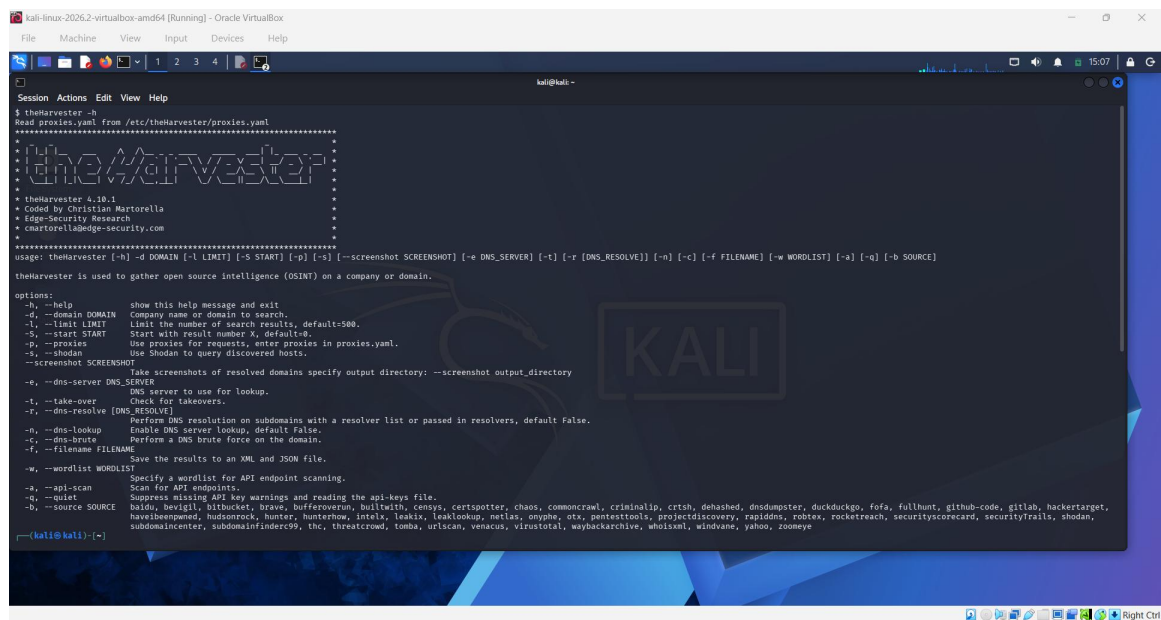
5.4.1 Task 1: Baidu Search

The assigned command was:
theHarvester -d microsoft.com -l 1000 -b baidu

The command used:

- -d for the target domain
- -l for the result limit
- -b for the selected source

Evidence

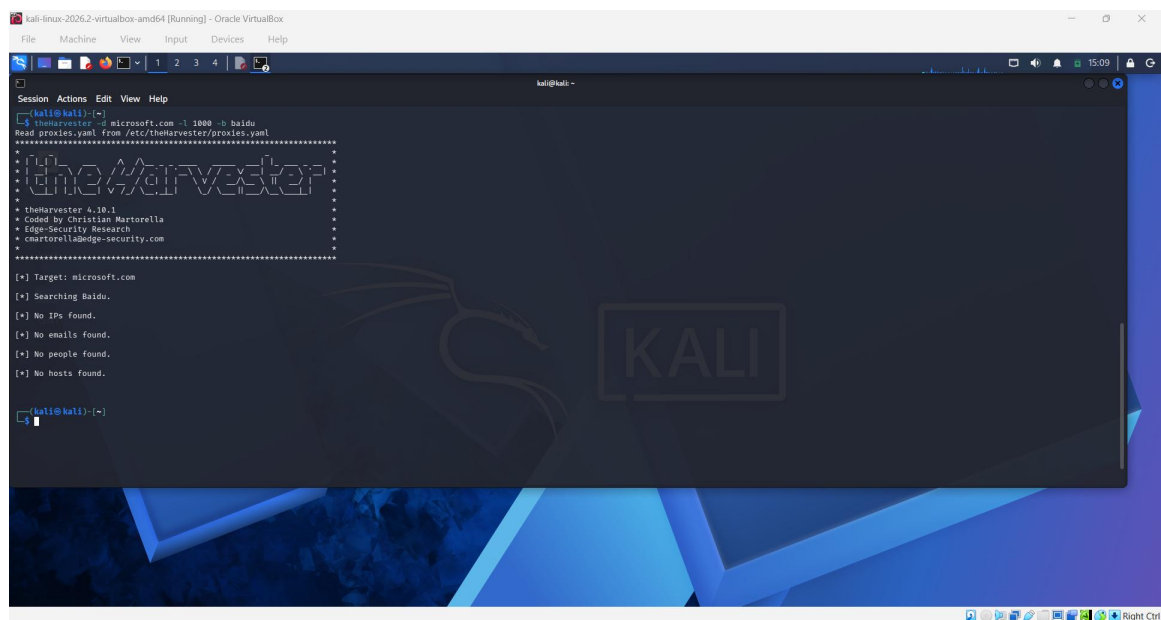


```
kali@kali:~$ theHarvester -h
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
theHarvester
*****
theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****
usage: theHarvester [-h] -d DOMAIN [-l LIMIT] [-s START] [-p] [-s] [--screenshot SCREENSHOT] [-e DNS_SERVER] [-t] [-r [DNS_RESOLVE]] [-n] [-c] [-f FILENAME] [-w WORDLIST] [-a] [-q] [-b SOURCE]

theHarvester is used to gather open source intelligence (OSINT) on a company or domain.

options:
  -h, --help            show this help message and exit
  -d, --domain DOMAIN    Company name or domain to search.
  -l, --limit LIMIT       Limit the number of search results, default=500.
  -s, --start START       Start with result number X, default=0.
  -p, --proxies           Use proxies for requests, enter proxies in proxies.yaml.
  -s, --shodan           Use Shodan to query discovered hosts.
  --screenshot SCREENSHOT Take screenshots of resolved domains specify output directory: --screenshot_output_directory
  -e, --dns-server DNS_SERVER DNS server to use for lookup.
  -t, --take-over         Check for takeovers.
  -r, --dns-resolve [DNS_RESOLVE] Perform DNS resolution on subdomains with a resolver list or passed in resolvers, default False.
  -n, --dns-lookup        Enable DNS server lookup, default False.
  -c, --dns-brute         Perform a DNS brute force on the domain.
  -f, --filename FILENAME Save the results to an XML and JSON file.
  -w, --wordlist WORDLIST Specify a wordlist for API endpoint scanning.
  -a, --api-scan          Scan for API endpoints.
  -q, --quiet             Suppress missing API key warnings and reading the api-keys file.
  -b, --source SOURCE     baidu, Bing, bitbucket, brave, bufferoverrun, builtwith, censa, certspotter, chaos, commoncrawl, criminalip, crtsh, dehashed, dnsdumpster, duckduckgo, fofa, fullhunt, github-code, gitlab, hackertarget,
                        haveibeenpwned, hudsonrock, hunter, hunterhow, intelx, leakls, leaklookup, metasploit, onghive, otx, pentesttools, projectdiscovery, rapid7, robtex, rocketreach, securityscorecard, securitytrails, shodan,
                        subdomaincenter, subdomainfinder99, thec, threatcrowd, tomba, urlscan, venacius, virustotal, waybackarchive, whoisxml, windvane, yahoo, zoomeye
```

Figure 15. theHarvester reconnaissance



```
kali@kali:~$ theHarvester -d microsoft.com -l 1000 -b baidu
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
theHarvester
*****
theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****

[*] Target: microsoft.com
[*] Searching Baidu.
[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] No hosts found.
```

Figure 16. theHarvester Task 1 result

Security Relevance

The exercise demonstrated how a single public information source can contribute to domain reconnaissance.

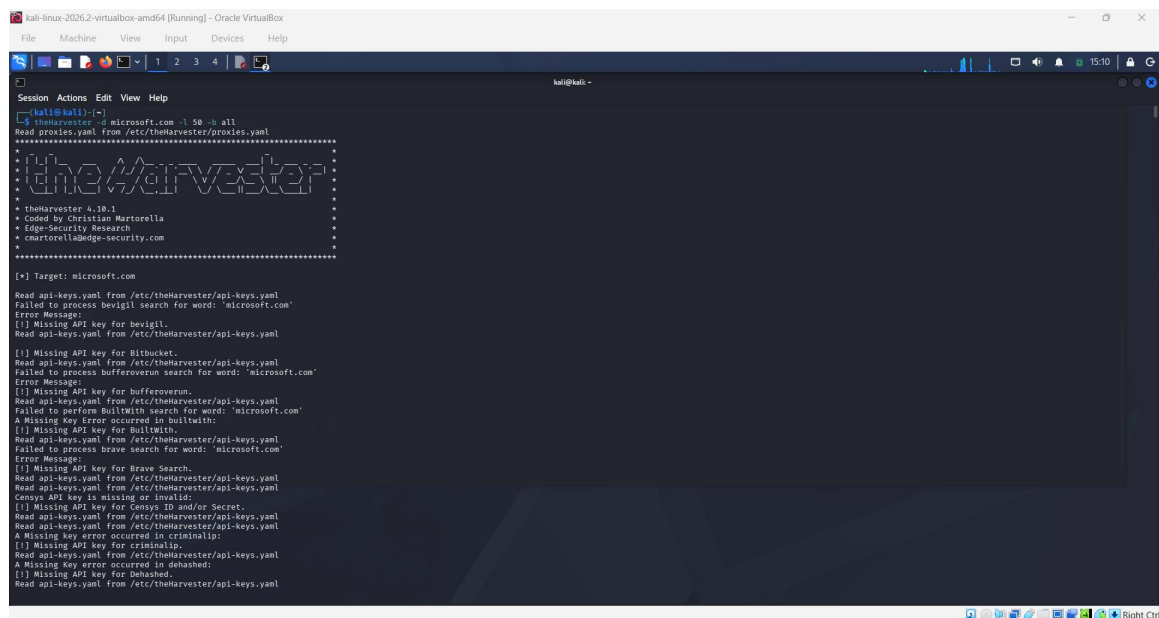
5.4.2 Task 2: All Sources

The assigned command was:

theHarvester -d microsoft.com -l 50 -b all

The command used the available sources supported by the installed version of theHarvester.

Evidence

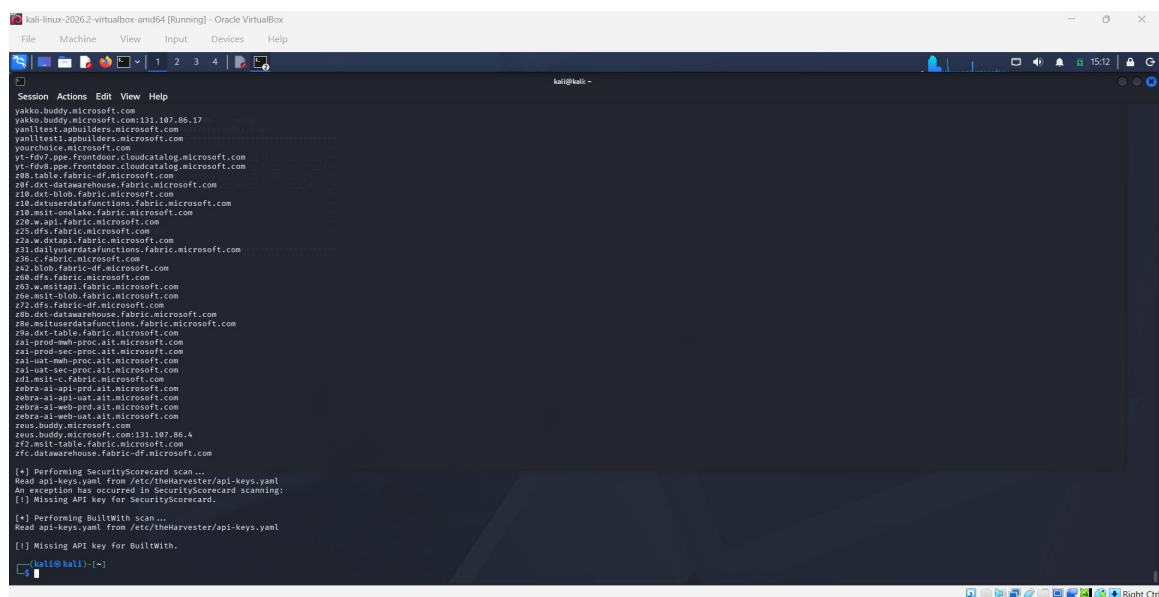


```
kali@kali:~$ theHarvester -d microsoft.com -l 50 -b all
theHarvester
theHarvester 4.10-1
  * Coded by Christian Martorella
  * EdgeSecurity Research
  * cartonnell@edge-security.com
  *

[*] Target: microsoft.com

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverun search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bufferoverun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform bullwith search for word: 'microsoft.com'
A Missing key error occurred in bullwith:
[!] Missing API key for Bullwith.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process brave search for word: 'microsoft.com'
Error Message:
[!] Missing API key for Brave Search.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Censys API key is missing or invalid:
[!] Missing API key for Censys ID and/or Secret.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing key error occurred in criminalip:
[!] Missing API key for criminalip.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing key error occurred in dehashed:
[!] Missing API key for Dehashed.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
```

Figure 17. theHarvester all-source scan



```
kali@kali:~$ theHarvester -d microsoft.com -l 50 -b all
yakko.buddy.microsoft.com
yakko.buddy.microsoft.com:131.107.86.17
yanlltest.azurebuilders.microsoft.com
yanlltest.azurebuilders.microsoft.com
yourchoice.microsoft.com
vt-fdv7.ppe.frontdoor.cloudcatalog.microsoft.com
vt-fdv8.ppe.frontdoor.cloudcatalog.microsoft.com
x86-table.fabric.df.microsoft.com
20f.dxt-datawarehouse.fabric.microsoft.com
210.dxt-blob.fabric.microsoft.com
210.dxtunidatafunctions.fabric.microsoft.com
210.msi-table.fabric.microsoft.com
220.w.api.fabric.microsoft.com
225.dfs.fabric.microsoft.com
225.w.datapi.fabric.microsoft.com
231.dailyuserdatafunctions.fabric.microsoft.com
230.c.fabric.microsoft.com
242.blob.fabric.df.microsoft.com
280.dfs.fabric.microsoft.com
280.w.msi-tag.fabric.microsoft.com
286.msi-blob.fabric.microsoft.com
272.dfs.fabric.df.microsoft.com
280.dxt-datawarehouse.fabric.microsoft.com
286.msiunidatafunctions.fabric.microsoft.com
29a.dxt-table.fabric.microsoft.com
2a1-prod-muh-proc.ait.microsoft.com
2a1-prod-sec-proc.ait.microsoft.com
2a1-uat-muh-proc.ait.microsoft.com
2a1-uat-sec-proc.ait.microsoft.com
2d1.msi-c.fabric.microsoft.com
2e8ra-ai-api-prod.ait.microsoft.com
2e8ra-ai-api-uat.ait.microsoft.com
2e8ra-ai-web-prod.ait.microsoft.com
2e8ra-ai-web-uat.ait.microsoft.com
zeus.buddy.microsoft.com
zeus.buddy.microsoft.com:131.107.86.4
2f2.msi-table.fabric.microsoft.com
2fc.datawarehouse.fabric.df.microsoft.com

[*] Performing SecurityScorecard scan...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
An exception has occurred in SecurityScorecard scanning:
[!] Missing API key for SecurityScorecard.

[*] Performing Bullwith scan...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for Bullwith.
```

Figure 18. theHarvester Task 2 output

Security Relevance

Using multiple sources demonstrates how information from different public platforms can contribute to a larger reconnaissance profile.

TheHarvester results may change over time because public sources, search indexes and APIs change.

5.5 W2-PM5: NETWORK SCANNING WITH ZENMAP

W2-PM5 focused on authorized network discovery using Zenmap.

The scan was performed against the tester's own local LAN.

5.5.1 Task 1: Install and Configure Zenmap

Zenmap was used as the graphical interface for Nmap-based network discovery.

The tool was configured for the local-network scanning exercise.

5.5.2 Task 2: Identify Local IP and LAN Subnet

The local Windows network configuration was used to determine the IP address and subnet applicable to the scanning exercise.

Evidence

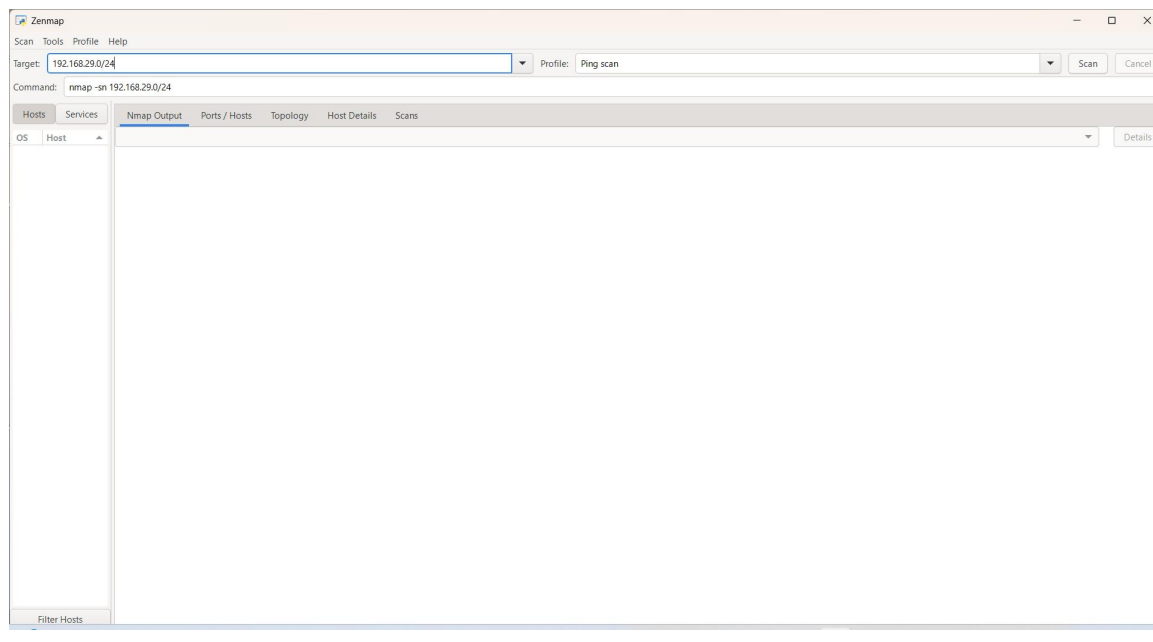


Figure 19. Local network configuration / Zenmap setup

5.5.3 Task 3: Find Live Hosts

Zenmap was configured to perform a Ping Scan against the authorized local LAN subnet.

The scan identified active hosts on the local network.

Evidence

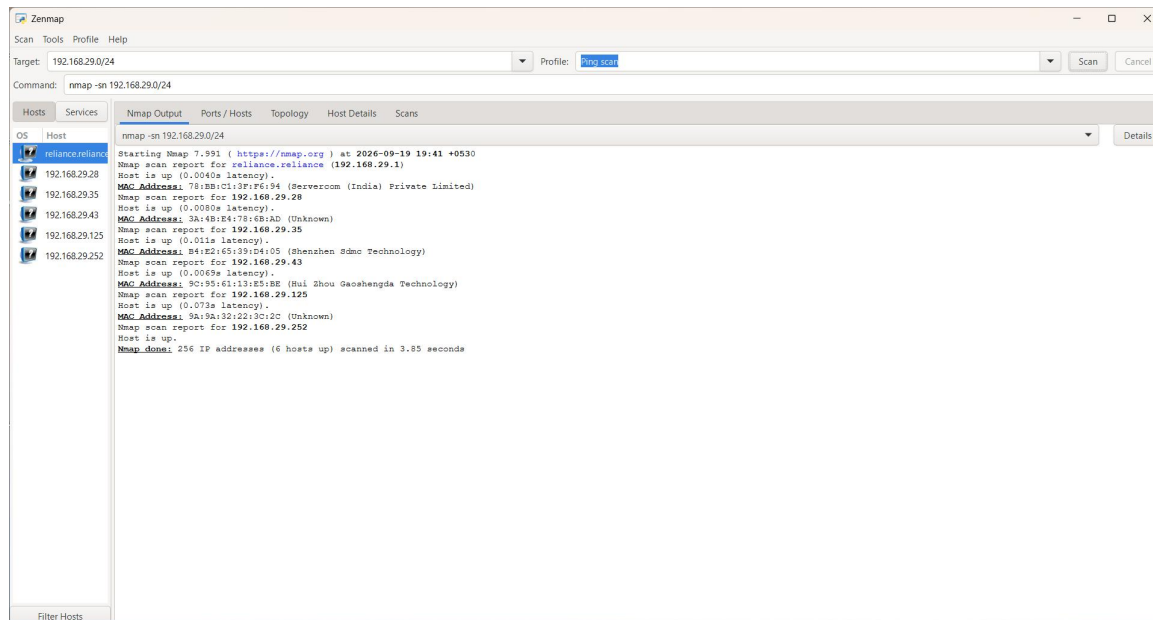


Figure 20. Zenmap live-host discovery

Security Relevance

Live-host discovery helps establish which devices are currently active on a network.

5.5.4 Task 4: Count Live Hosts

The number of live hosts was determined from the Zenmap scan results.
The complete scan output was retained as evidence.

Answer : 6 hosts up.

5.5.5 Task 5: Identify IP Addresses

The IP addresses of discovered live hosts were reviewed from the Zenmap results.

Answer :

Servercom (India) Private Limited: 78:BB:C1:3F:F6:94

Unknown: 3A:4B:E4:78:6B:AD

Shenzhen SDMC Technology: B4:E2:65:39:04:05

Hui Zhou Gaoshengda Technology: 90:95:61:13:E5:BE

Unknown: 9A:9A:32:22:30:2C

5.5.6 Task 6: Identify MAC Addresses

MAC addresses associated with discovered devices were reviewed from the authorized local-network scan.

Because MAC addresses are device identifiers, they should not be unnecessarily published in a public repository.

Evidence

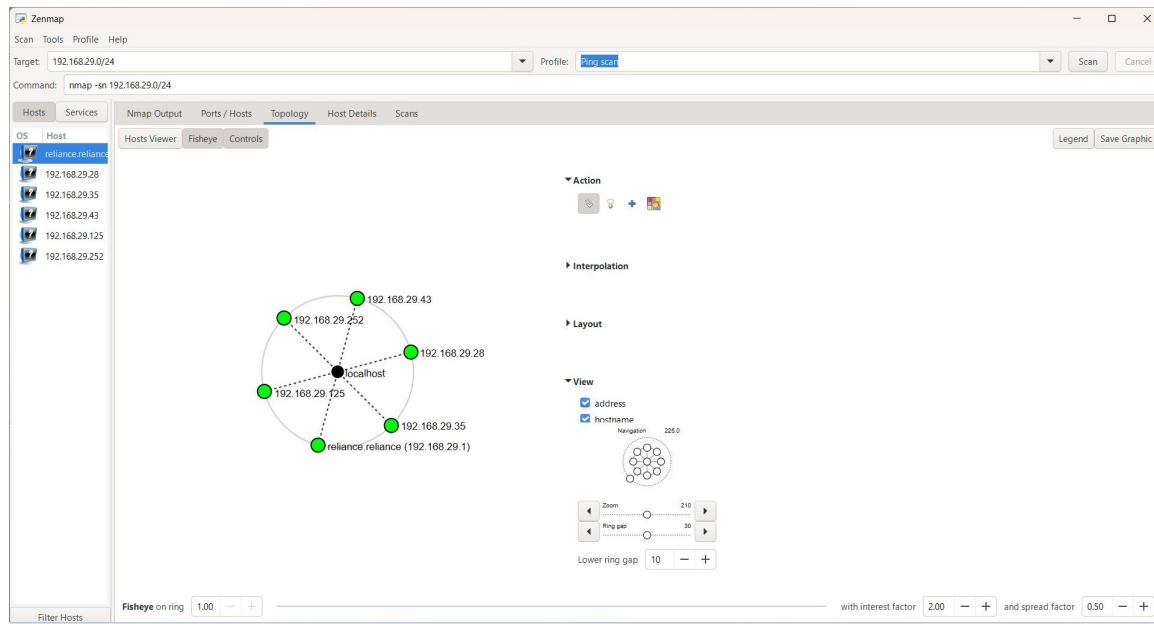


Figure 21. Fisheye

5.5.7 Task 7: Generate Network Topology

Zenmap's topology feature was used to visualize the discovered local network. The topology was saved as a PDF.

Evidence

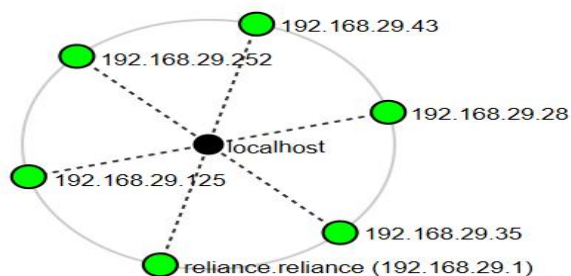


Figure 22. Zenmap network topology

Security Relevance

Network topology discovery helps security professionals maintain an accurate inventory of active devices and understand the structure of a local network.

6. FINDINGS AND OBSERVATIONS

#	Observation	Module	Security Relevance
1	Domain registration and DNS information was publicly available	W2-PM1	Supports infrastructure profiling
2	Web technologies were identifiable	W2-PM1	Provides technology fingerprinting information
3	Public DNS resolution exposed infrastructure information	W2-PM1	Assists network reconnaissance
4	HTTP response headers exposed technical information	W2-PM1	May assist application fingerprinting
5	WAF technology was identifiable	W2-PM1	Reveals part of defensive architecture
6	Multiple DNS records were publicly observable	W2-PM1	Helps construct an infrastructure profile
7	Search-engine indexed resources were discoverable	W2-PM2	Demonstrates potential public-information exposure
8	Public-source information could be correlated	W2-PM3	Supports organizational profiling
9	Multiple public sources could be queried for reconnaissance	W2-PM4	Demonstrates information aggregation
10	Active devices were discoverable on the local LAN	W2-PM5	Supports internal asset discovery

These observations should not be interpreted as confirmed vulnerabilities. No exploitation or vulnerability validation was performed.

7. RISK ANALYSIS AND IMPACT

Observation	Potential Impact	Suggested Priority
Public technology information	Technology fingerprinting	Medium
Public DNS information	Infrastructure profiling	Low-Medium
HTTP technical information	Application fingerprinting	Low
Search-indexed resources	Unintended information exposure	Medium
Public organizational information	Reconnaissance and profiling	Medium
Discoverable local devices	Asset visibility	Medium

The priorities above describe the potential security relevance of the observations, not confirmed vulnerabilities.

8. RECOMMENDATIONS

- Regularly review publicly exposed technologies and services.
- Keep CMS platforms, plugins and libraries updated.
- Review HTTP response headers for unnecessary technical information.
- Periodically audit public DNS records.
- Maintain and monitor WAF configuration.
- Review search-engine indexing for sensitive or unintended resources.
- Monitor publicly exposed organizational information.
- Maintain an accurate internal asset inventory.
- Periodically perform authorized internal network discovery.
- Investigate unknown devices discovered during internal scans.
- Keep network topology documentation updated.
- Perform all security testing within an approved and documented scope.

9. LEARNING OUTCOMES

The Week 2 practical activities provided experience in:

- Domain footprinting
- WHOIS analysis
- Web technology fingerprinting
- DNS reconnaissance
- HTTP header analysis
- WAF identification
- DNS record enumeration
- Search-engine reconnaissance
- Maltego-based information correlation
- Public-source reconnaissance
- theHarvester usage

- Local network discovery
- Live-host identification
- IP address identification
- MAC address identification
- Network topology generation
- Security evidence collection
- Technical report writing
- Ethical and authorized security testing

10. ETHICAL AND LEGAL CONSIDERATIONS

All activities were conducted for educational purposes and within the assigned scope.

No attempt was made to:

- Bypass authentication
- Obtain unauthorized credentials
- Exploit vulnerabilities
- Escalate privileges
- Perform denial-of-service activity
- Modify or delete data
- Access private information
- Scan systems outside the authorized scope

The Zenmap exercise was restricted to the tester's own local LAN.

Sensitive information such as cookie values and unnecessary device identifiers should be redacted before publishing evidence in a public repository.

11. CONCLUSION

Week 2 of the Networkwalks Cybersecurity & Ethical Hacking Internship was completed through five practical modules covering reconnaissance, footprinting and network discovery.

W2-PM1 provided practical experience with WHOIS, WhatWeb, NSLookup, cURL, Wafw00f and DNSRecon. These tools demonstrated how domain registration, DNS infrastructure, web technologies, HTTP information and defensive technologies can be identified during reconnaissance.

W2-PM2 demonstrated search-engine reconnaissance using GHDB techniques and showed how publicly indexed resources can be identified through carefully constructed search queries.

W2-PM3 introduced Maltego for graphical reconnaissance and information correlation, including domain-based email reconnaissance.

W2-PM4 demonstrated public-source reconnaissance using theHarvester against the assigned domain and multiple information sources.W2-PM5 provided practical experience with Zenmap for authorized local network discovery, including live-host identification, IP and MAC address identification, and topology generation.

The combined exercises demonstrated that reconnaissance can reveal significant information before any exploitation phase begins. They also reinforced the importance of authorization, scope control, responsible evidence handling and accurate technical documentation.The Week 2 objectives for reconnaissance, footprinting and network discovery were completed successfully.

12. REFERENCES

Networkwalks Training Materials

- W2-PM1: Footprinting
- W2-PM2: Footprinting & Reconnaissance with GHDB
- W2-PM3: Footprinting with Maltego
- W2-PM4: Footprinting & Reconnaissance with theHarvester
- W2-PM5: Network Scanning with Zenmap

Technical References

- Nmap Project documentation
- Kali Linux documentation
- Maltego documentation
- theHarvester documentation
- WhatWeb documentation
- Wafw00f documentation
- DNSRecon documentation
- ICANN / WHOIS resources
- Google Hacking Database reference material

The references were used to understand the tools and techniques. The actual observations documented in this report are based on the practical exercises and collected repository evidence.

13. AUTHOR

Ishanya Jha
Cybersecurity & Ethical Hacking Intern
B083-Networkwalks
Networkwalks Cybersecurity Internship
Week 2 Project Report

14. PROJECT INFORMATION

Field	Details
Program	Cybersecurity & Ethical Hacking Internship
Organization	Networkwalks
Week	Week 02
Report	W2-PM-FINAL
Modules	W2-PM1, W2-PM2, W2-PM3, W2-PM4, W2-PM5
Primary Focus	Footprinting, Reconnaissance & Network Scanning
Phase 1	Completed
Phase 2	Completed
Overall Status	Week 2 Completed

END OF REPORT